

# Thomas VAUDESCAL

Quantitative Developer | Quantitative Researcher

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## Profile

Quantitative developer with an MSc in Financial Engineering from HEC Montréal (**GPA 3.88/4.3**). Builds backtesting, pricing and trading systems in **Python, C++ and Rust**, and takes strategies from research to live trading. Founder of Synerqo, previously at National Bank Financial Markets and Deloitte.

## Professional Experience

### Synerqo

Montreal, Canada

Founder & Quantitative Developer

May 2025 - Present

- Built an **event-driven options backtesting engine in Python and Numba** that models spreads, slippage, commissions, margin and assignment. It runs five years of SPX intraday option chains (**216M rows**) in **under a minute** on one workstation, with 2,448 automated tests.
- Designed, backtested and **deployed a systematic SPX options strategy to live trading**, turning MSc research on dealer gamma exposure into a production signal and execution workflow.
- Built an **options and structured products pricing engine**: Monte Carlo, PDE solvers, Heston, SABR and SVI calibration, and Greeks via **adjoint algorithmic differentiation (AAD)**, 2.9x faster than bump-and-reprice.
- Developed **Rust execution components** for market-data ingestion, quote updates and order routing, and profiled their latency against the Python research stack.
- Delivered client software: **Ganesha**, a retrieval-augmented (RAG) AI assistant in production, and a custom CRM for an insurance client that **saves it tens of thousands of dollars**.
- **Technologies**: Python, Numba, JAX, C++, Rust, FastAPI, PostgreSQL, TypeScript, QuantLib, QuantConnect.

### Deloitte

Montreal, Canada

Quantitative Analyst | Financial Engineering & Modeling - Financial Risk

October 2024 - June 2025

- Developed and validated **credit risk models (PD, LGD, EAD)** for major Canadian financial institutions under IFRS 9 and Basel III/IV frameworks.
- Led **independent model validation engagements**, assessing assumptions, calibration and discriminatory power through statistical backtesting, sensitivity analysis and stress testing.
- Built **automated validation pipelines and monitoring dashboards in Python**, cutting manual review time by **40%**.
- Authored technical validation reports and regulatory documentation, and presented findings to **senior stakeholders and client risk committees** under OSFI requirements.
- **Technologies**: Python, R, SQL, statistical modeling, model monitoring and validation.

### National Bank Financial Markets

Montreal, Canada

Quantitative Developer Intern | Global Equity Derivatives R&D

May 2023 - December 2023

- Built an **end-to-end Python backtesting framework**: signal generation, portfolio construction, transaction costs, risk analytics and performance attribution.
- Developed and deployed low-volatility and momentum **SmartBeta strategies covering \$800M in AUM** within a \$7B structured products division, with constrained portfolio optimization in cvxpy.
- Delivered **\$1M in annual cost savings** through strategy optimization and migration from an external vendor to internal production systems.
- Worked with **structuring and trading teams** to integrate quantitative signals into derivatives pricing and hedging workflows.
- Delivered performance and risk reports to the **trading desk and portfolio managers**.
- **Technologies**: Python, cvxpy, SQL, Bloomberg Terminal & API, AWS (EC2, S3), statistical testing.

### Videns Analytics

Montreal, Canada

Data Analyst

April 2020 - June 2021

- Led **insurance analytics consulting engagements**: data cleaning, predictive modeling and automated reporting.
- Designed a **Python, statistics and machine learning curriculum** adopted by UQAC for a university course. Promoted from intern to full-time.

Technical Skills

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|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming      | Python, C++, Rust, SQL, R, MATLAB, TypeScript.                                                                                                                                                  |
| Computer Science | Object-oriented design, design patterns, data structures and algorithms, event-driven architecture, concurrency (asyncio), performance optimization, latency profiling.                         |
| Python Stack     | NumPy, Numba, JAX, pandas, Polars, SciPy, scikit-learn, PyTorch, cvxpy, QuantLib.                                                                                                               |
| Quant Finance    | Backtesting, walk-forward validation, derivatives pricing, Greeks via AAD, Monte Carlo, PDE methods, volatility calibration (Heston, SABR, SVI), portfolio optimization, market microstructure. |
| Research Methods | Pre-registered hypotheses, deflated Sharpe ratio, probability of backtest overfitting, bootstrap, statistical testing.                                                                          |
| Risk Modeling    | Credit risk (PD, LGD, EAD), model validation, stress testing, model monitoring.                                                                                                                 |
| Tools            | Git, Linux, Docker, AWS (EC2, S3), DuckDB, PostgreSQL, FastAPI, REST and WebSocket APIs, Bloomberg Terminal & API.                                                                              |
| AI & Web         | RAG, LLM integration, Qdrant, React, Next.js.                                                                                                                                                   |

Education

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HEC Montreal

Montreal, Canada

MSc in Financial Engineering (Thesis) | GPA 3.88/4.3

August 2022 - December 2024

- **Thesis:** Gamma Exposure and Intraday Volatility Dynamics in the SPX Options Market. Graded **A+ with commendation from the examination committee**, now a working paper for the International Review of Financial Analysis. Applied to a live systematic options strategy.
- **Scholarships of excellence:** Deloitte M.Sc., Alpha Fixe Capital, Jocelyne & Jean C. Monty I and II.
- **Distinctions:** National Bank challenge winning team at ConUHacks VIII (2024), top 10 of 66 teams at the McGill FIAM Portfolio Management Hackathon, 2nd place at the HEC Montreal Trading Competition.
- **Leadership:** Vice President Research at the Trading Club and R&D Analyst at the BNI-HEC Investment Fund (\$5M AUM).
- **Selected coursework:** Stochastic Calculus, Derivatives Pricing, Financial Econometrics, Numerical Methods, Machine Learning for Finance, Risk Management.

HEC Montreal

Montreal, Canada

BBA | Mathematics, Finance and Economics

August 2017 - May 2022

- **GPA:** 3.8/4.3 in specialization courses. Quantitative track combining mathematics, economics and finance.
- **Leadership & teaching:** Vice President Education at the Data Science Committee (Python workshops, machine learning seminars), Teaching Assistant for Intermediate Microeconomics (200+ students).

University of Montreal

Montreal, Canada

Minor in Mathematics

August 2018 - August 2022

- Probability Theory, Mathematical Statistics, Stochastic Processes, Linear Algebra, Real Analysis, Programming (C/C++).

Languages

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|---------|-----------------------------------------------|
| French  | Native                                        |
| English | Professional proficiency (spoken and written) |

Interests

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|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technology   | Personal quant projects on GitHub: quant-paper-lab (finance papers reproduced on public data), fx_strategies (walk-forward framework), options market-data pipelines. |
| Intellectual | <b>Chess (2000 Elo)</b> , mathematics, philosophy.                                                                                                                    |
| Sports       | Track and field, marathon running, mountain hiking, indoor climbing, swimming, weight training, scuba diving.                                                         |